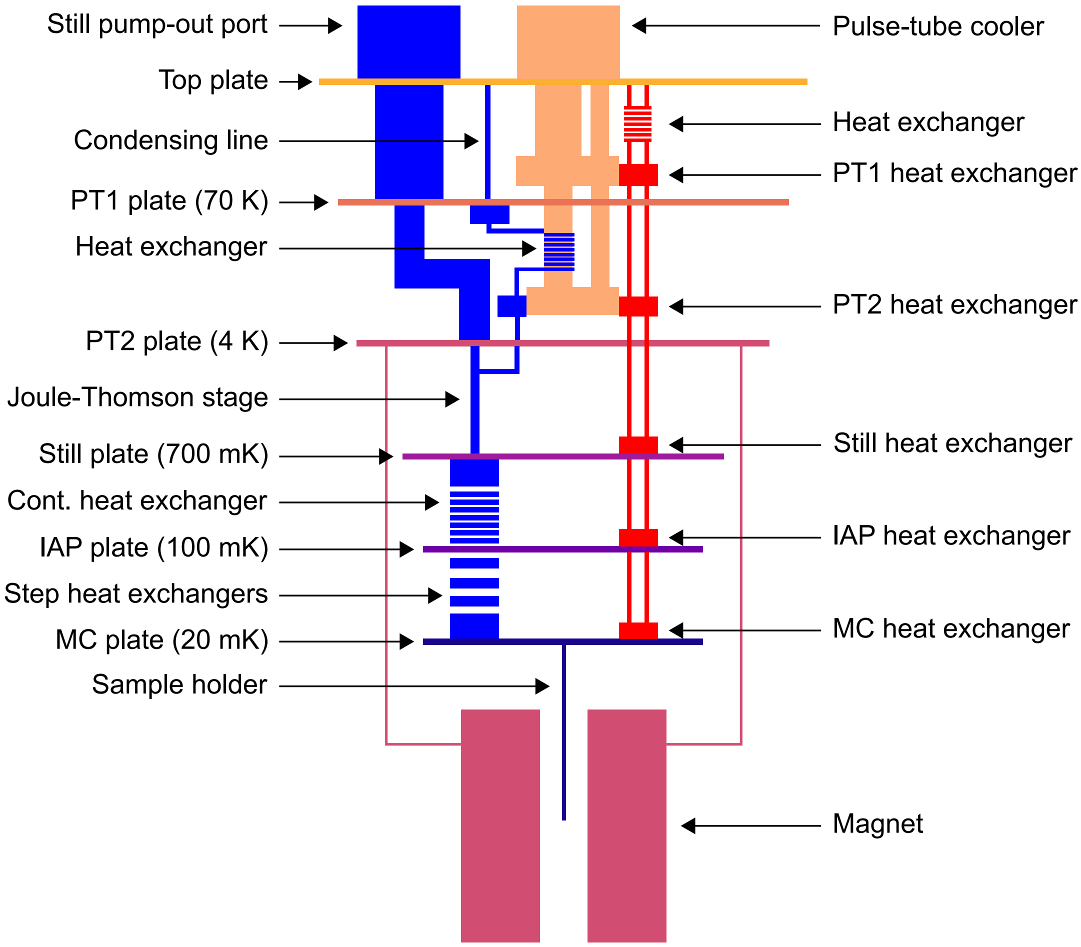

Figure 3.7: The Triton 200 schematics. The pulse-tube cooler is shown in orange, the pre-cool circuit in red, and the dilution circuit in blue. The different plates and their temperatures are also shown. The magnet (violet) is heated by the PT2 plate. Adapted from the Triton User Manual, courtesy of Oxford Instruments.

The DC electronic system makes use of a low-noise home-made equipment (Delft Box), which is controlled using the Python interface QTlab (Fig. 3.8).

Digital values from a desktop computer are sent into an optical signal generator, which produces a serial input for the digital-to-analog converter (DAC), which generates and reads voltages (bias V_{SD} and V_{G}). The four accessible DACs each have a 16-bit resolution and can read/write voltages between $\pm 2\,\mathrm{V}$. Higher voltages are managed by using a high-impedance voltage source. The entire unit is battery-powered to provide complete isolation from the main source of electronic noise, and the switch box connected to the Delft box provides simple switching between

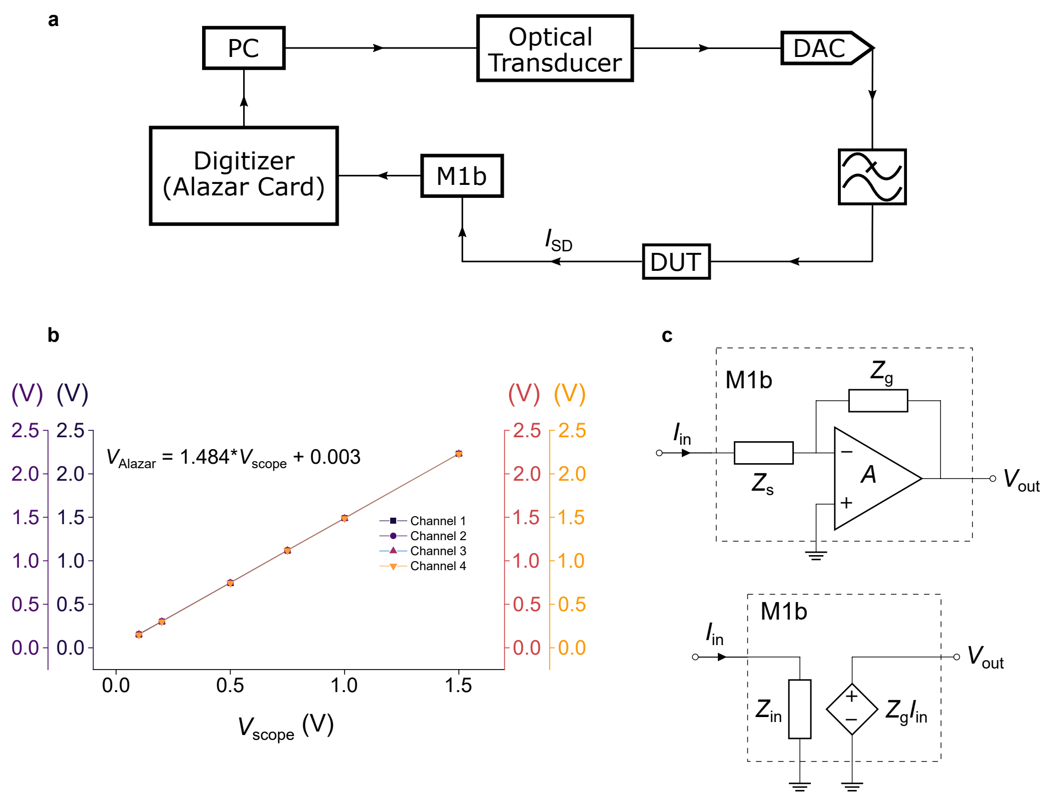

Figure 3.8: **a,** Schematic view of the DC electronics setup used in cryogenic transport. The low-pass filters before the DUT encompass RC with cutoff frequencies f_T between 5–11×10^3 Hz, Cu powder ($f_T \simeq 100$ kHz), and LC filters. **b,** The Alazar card four channels have been calibrated against known voltages output from an oscilloscpe (scope for short). **c,top** Current-to-voltage simplified schematic and its equivalent circuit **c,bottom**. For the M1b adopted in our setup, a manual dial allows changing Z_g between 1–1×10^3 MΩ.

different devices as well as critical protection against electronic discharge (ESD) that could permanently damage the sample.

Input signals from the switch board are routed through the internal circuitry of the Delft box before entering the fride. After passing through a series of low-pass filters, the current is cooled by two sapphire heat sinks on the PT2 plate and two on the MC plate. An RC filter, a meandering-line copper powder filter, three LC filters, and another RC filter are included. The input signals are then routed to the sample through a 48-wire NanoD connection. The sample chip is wire-bonded to a 48-pin chip holder, which is mounted on a home-made printed circuit board.

The DUT's source-drain current travels in reverse order via the specified components until it reaches the current amplifier (Fig. 3.8), which can amplify and monitor currents with noisefloors as low as $5\,\mathrm{fA\,Hz^{-0.5}}$ (at $1\,\mathrm{GV\,A^{-1}}$). The amplified output is then supplied into the Alazar data acquisition board via a coax wire, with the sampling rate set to 100 $\mathrm{MS^{-}1}$ throughout the most portion of our experiment. The QTLab interface specifies an integration time t_{int} during which the Alazar card samples the source-drain current before computing and storing the average.

It is necessary to emphasize that the voltages read by the Alazar card are inaccurate. This is likely due to the input impedance setting, which causes the voltage to drop. To overcome this, a calibration curve (Fig. 3.8b) was constructed by measuring known voltages with both the Alazar card and an oscilloscpe. All the numerical data presented in this thesis have been rescaled by using analogous calibration curves.

4

Instrumentation Development

The instrumentation described in the previous chapter was based on the work of former members of my team. In this chapter, I describe the further enhancements and setup I have partially or entirely accomplished during the course of my thesis. In this regard, I will first describe the measuring technique I implemented in the cryogenic setup to ensure accurate results even beyond the cryostat heaters' operating temperature range. Second, I will describe the installation and testing of micro Hall sensors utilised in magnetometry measurements.

4.1 Temperature-dependent Protocol

Dilution refrigerators are suited for cryogenic operations although it is possible to operate at higher temperature by working with still and mixing chamber heaters. In order to measure reliably up to room temperatuer, however, one must ensures that the temperature drift is ideally slower than the time taken to acquire experimental measurements. Since there is no well-defined experimental procedure for doing so, a workaround which leverages the Triton system was found in this thesis.

Triton heaters are capable to control the temperature precisely from base temperature ($\sim$ 25 mK) up to 30 K. In order to stabilise the temperature above this range, one has to generate a thermal link between the sample and the environment.

First the heaters (mixing chamber and still, see Ch. 2) are switched off: this enables main heat sources to be excluded from exchanging thermal energy with the sample inside the mixing chamber. Being the mixing chamber very well thermally isolated, the temperature will start raising very slowly: to give a reference number, ∼24h after the heaters were switched off the temperature rose by only 2 K. The pre-cooling loop is then filled with the He3/He4 mixture, creating a thermal link between the various stages within the dilution refrigerator and the mixing chamber. This results in a faster heating rate, but slow enough to allow to take quick measurements such as $I - V_{\mathrm{SD}}$ and $I - V_{\mathrm{G}}$ traces.

Triton heaters are able to precisely control temperatures from the base temperature up to 30 K. To stabilise the temperature above this range, one must establish a thermal connection between the sample and the surrounding environment. The mixing chamber and still heaters are turned off initially. This process prevents the primary heat sources from exchanging thermal energy with the sample inside the mixing chamber.

Due to the mixing chamber's excellent thermal isolation, the temperature will begin to rise extremely slowly: 24 hours after the heaters were turned off, the temperature rose from 30 K to 32 K by only 2 K. The pre-cooling loop is subsequently filled with the He_3/He_4 mixture, establishing a thermal connection between the dilution refrigerator's various phases and the mixing chamber. This causes a quicker heating rate, but one that is still slow enough to permit measurements such as bias and gate traces.

The Alazar card, Triton, and IVVI (Delft box modules) drivers are initialised and assigned global variables that maintain their state during the execution of the application. Afterwards, multiple variables are initialised. The precooled variables are a Triton driver-implemented process that enables direct instrument command setting and retrieval. While below 1 K, the RuO_2 thermometer is employed, but instead to accurately read temperatures over 1 K, the Cernox thermometer must be used; hence, the temperature channel variable is set to channel 5 within the while loop if this condition is met. The application then acquires measurements of the gate trace and stability diagram. The algorithm designed and implemented in Python language is detailed in Alg. 4.1

Algorithm 4.1 Triton High Temperature Measurement

```
Require: Initialise Alazar, Triton and IVVI
Ensure: Alazar, Triton, and IVVI are global
Initialise Final Temperature
Initialise Bias Array
Initialise Target Temperatures Array
Current Temperature ← Read Temperature
Temperature Step ← 3
Temperature Channel ← 8
Precooled ← False
while Current Temperature < Final Temperature do
    Precooled ← True
    T ← Read Temperature
    if T > 1.5 then
        Temperature Channel ← 5
    end if
    for length Bias Array do
        Run Gate Trace Measurement
    end for
    T ← Read Temperature
    if T is in Target Temperatures Array then
        Precooled ← False
        Run Stability Diagram
    end if
end while
```

4.2 AC Measurements

Towards the end of my transport experiments, I found a solution and integrated the lock-in amplifier into the existing Delft Box configuration. The last effort failed previously owing to an impedance mismatch when the Delft box output was fed into the alazar card directly, hampering the dection of any signal.

It was necessary to first wire the DAC module. The iso-amp output (S0) module is not coupled with the summing module by default. However, the pattern is easily rewritten by sampling and redecorating the wire according to the scheme. Thus, the Lock-in output was linked to the S0 Delft box and fed with the signal from the M0 module. The signal was then locked at the proper frequency and phase-adjusted before being supplied to the desktop computer via usb or ethernet. Experiment routines were readily built and integrated into the existing setup by instantiating

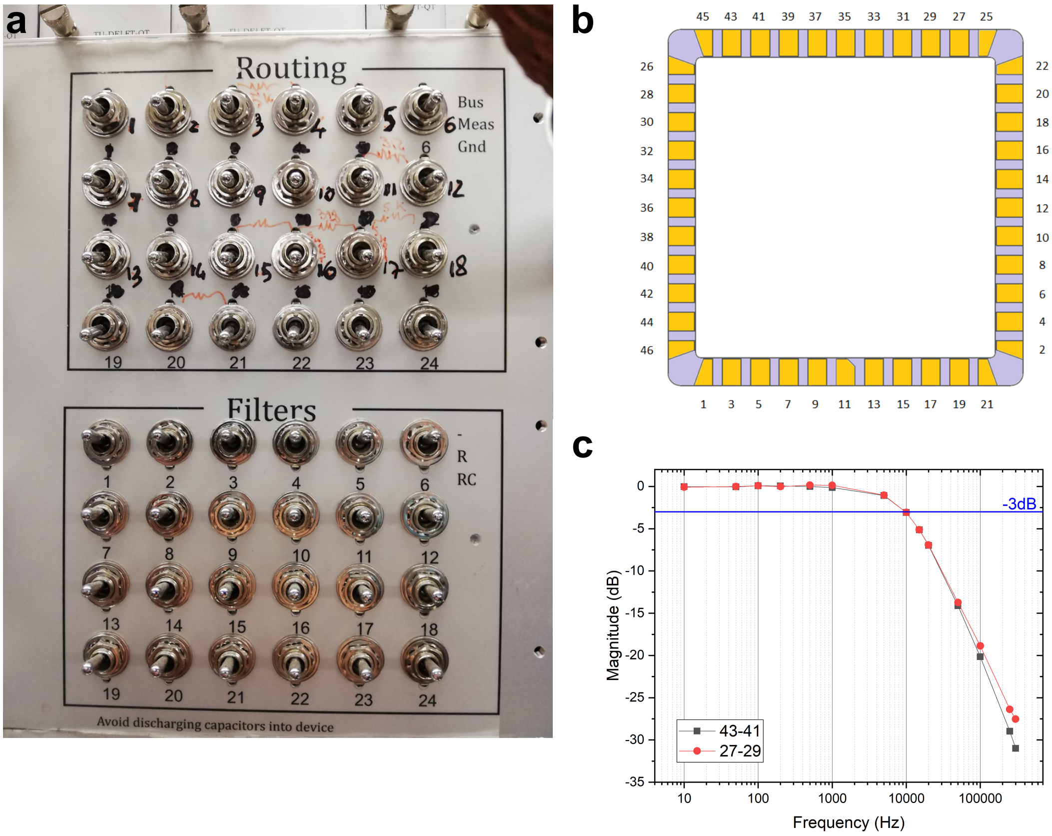

Figure 4.1: a, Breakout box of DC connections. A second breakout box is used for connections from 26 to 48. **b,** Chipholder schematic with each connection numbers: pin 11 is used to orientate the chip in the correct direction. **c,** Frequency test. Powder (Cu) and RC filters set frequency operation limit theoretically up to 10 kHz. In practice we experimentally observed a statistical significance degradation of the signal already at 5 kHz.

a lock-in instance and integrating it into the superclass station, which contains instances for all the other required instruments.

4.2.1 Integration of Lock-in Amplifier with Delft Box

A great deal of information about both the theory and tips of integrating a lock-in with and existing low-noise setup such as the one I have used may be found in the excellent reference [187]. Here I briefly report and comment salient points that I have employed to successfully integrate a standard SR-830 lock-in with Delft Box. Same concepts were later applied for integrating two Zurich Instruments lock-in amplifiers as well.

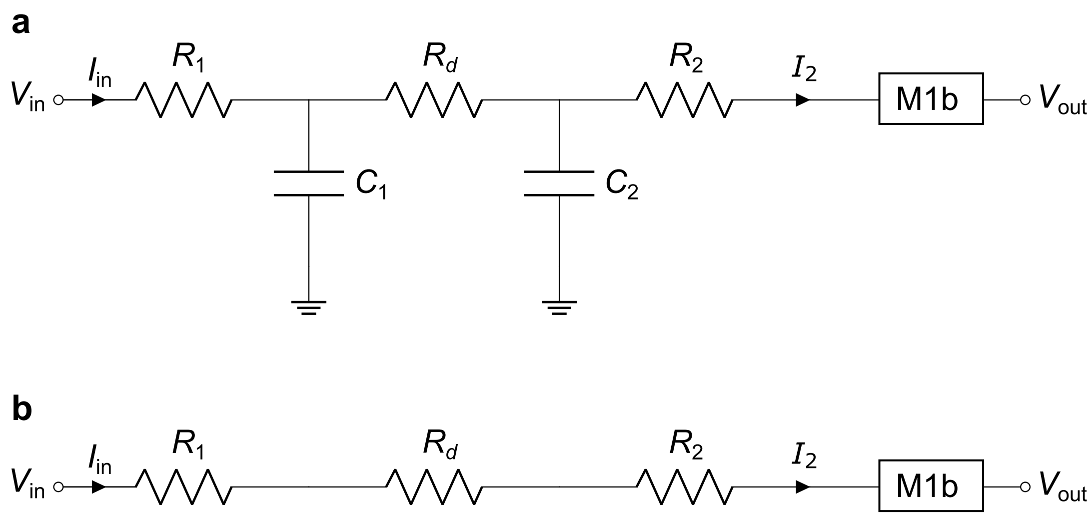

Figure 4.2: a, An example of circuit diagram of a voltage-biased sample R_d in series with two RC filters and measured by a current-to-voltage (M1b) converter. **b,** Because the device resistance is finite and the operating frequency is well below the DC lines bandwidth, it is assumed that capacitors' frequency response plays no role.

Although the measured cryogenic setup noisefloor level is a tenth of pA, the intrisic $1/f$ noise represents an instrinsic and important hurdle in any DC experiment. As we saw in Sec. 2.5.2, the central quantity in an electron transport experiment is the tunneling density of states, which is ultimately related to the derivative of the current in a $I - V_{\text{SD}}$ experiment. Numerically retrieving the conductance may introduce noise due to the non-negligible current fluctuations between two measurement points; this picture gets even further complicated if one takes into account possible charge jumps occuring during the acquisition.

The logic behind a lock-in is as surprinsingly simple as clever. The idea is to fed an oscillating voltage $V_{\text{osc}}(f)$ into the pure DC signal in order to produce an oscillating current $I_{\text{tot}}(f) = I_{\text{DC}} + I_{\text{AC}}(f)$ and then later to filter out the noise outside a narrow band $f \pm \tau^{-1}$ for a settable integration time τ (Fig. 4.3). As the reference at the top of this sections points out, lock-in conductance measurements can readily introduce errors on the scale of 10 points percent of the total signal if operated with the incorrect parameters settings, wherease DC is not susceptible

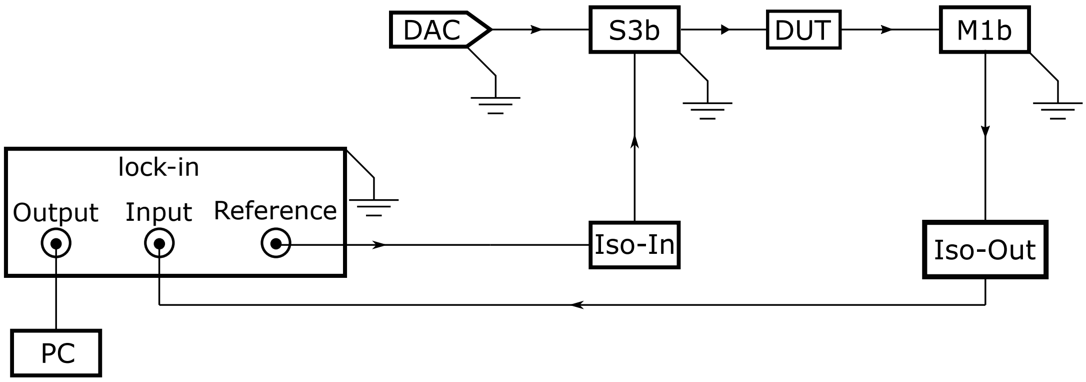

Figure 4.3: Alternating current (AC) electronics setup. The oscillating signal is superimposed on a DC-only signal originating from the DAC and sent into the S3b voltage/current source module through the iso-amplifier input. Note that a 0.01 factor is introduced by the S3b control input to allow the superposition of a frequency-modulated signal on top of a dc-sweep. [188].

to these complex-space-caused aberrations. Therefore, it is advisable to take extra caution if accurate conductance values are required from your lock-in measurement. More loose conditions might be consider, on the contrary, if features are desired rather than accurate values.

Fridge lines are filter with RC and Cu powder filter. At finite frequency $\omega = 2\pi f$, each circuit component response has to be modeled in the complex plane. As long as all the k capacitors active across all the stages within the fridge do not draw any current, i.e. $R_i \ll 1/(\omega C_k)$ where the subscript i denotes the number of resistance, it is safe to exclude any parasitic components when one attempts to recover the resistance value for the device under test.

This principle might be thought as the requirements to work with oscillating voltages way below that the cut-off frequency imposed by the filters, namely:

$$f \ll f_{\mathrm{cutoff}} = \frac{1}{2\pi R_i C_k} \tag{4.1}$$

Fig. 4.1c demonstrates the cut-off frequency set by out setup to be around 1 kHz, even though the -3 dB point lies around one order of magnitude higher. The finite-frequency efficiency and response of the DC lines is, however, not the only factor that one has to take into account when adopting AC techiniques.

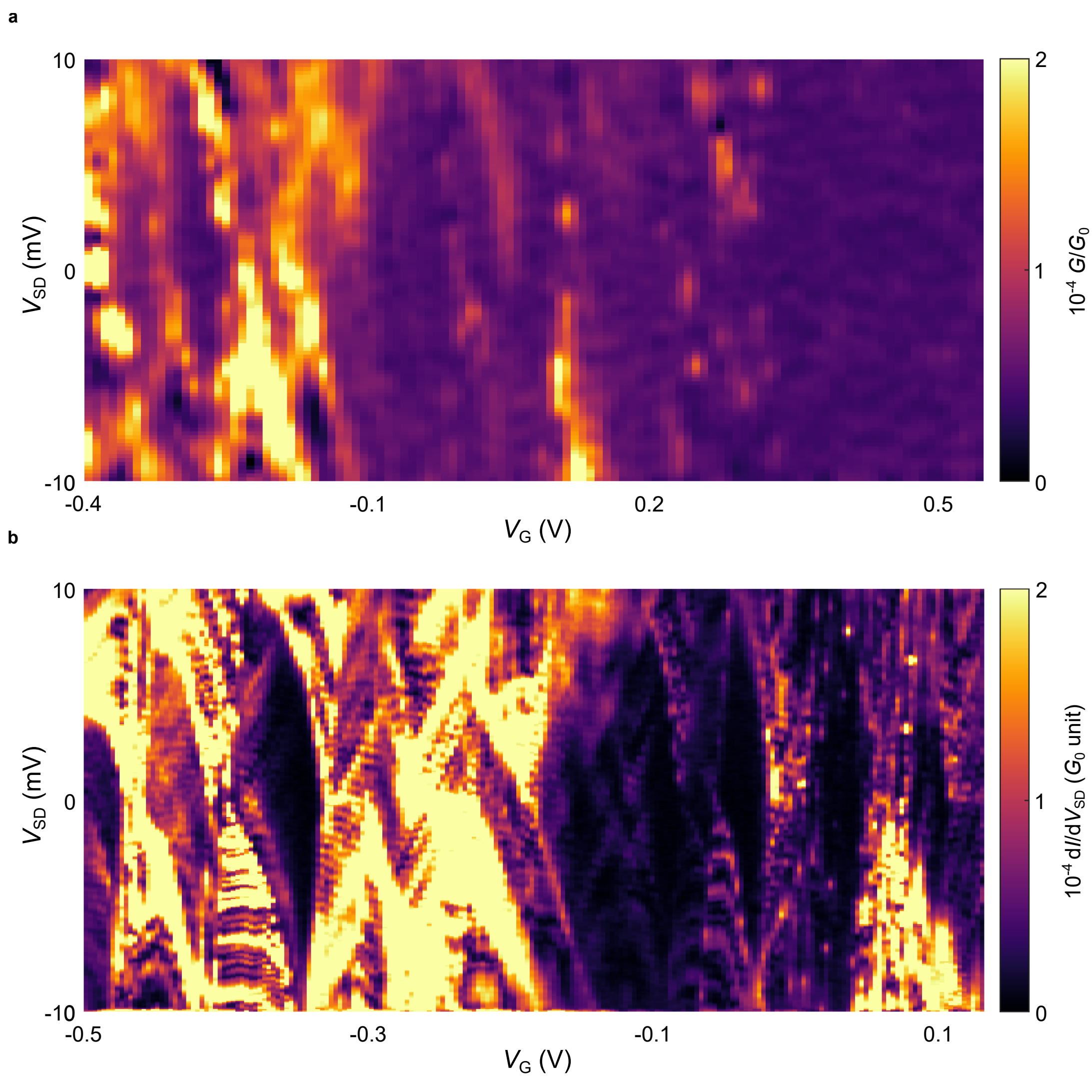

Figure 4.4: Comparison of **a,** DC and **b,** AC stability diagrams acquired for the same device at $T = 20\,\mathrm{mK}$. The lock-in excitation voltage felt by the device was $V_{\mathrm{osc}} = 100\,\mathrm{\mu V}$ whereas the sensistivity and the time constant were set at $200\,\mathrm{mV}$ and $\tau = 0.5\,\mathrm{s}$, respectively.

Due to its narrow bandwidth of approximately 200 kHz, the most commonly used M1b current amplifier in the IVVI rack is especially susceptible to lock-in finite-frequency problems. Slower amplifiers require less operating current, so the objective of this amplifier's design is to minimise its back action on the sample. Nonetheless, this constrains the operating frequency to approximately 30 kHz†. When this

†The Zurich Instruments lock-in contains a built-in functionality that allows frequency sweep while monitoring the voltage noise calculated in $\mathrm{nV}/\sqrt{\mathrm{Hz}}$ for a given lock-in measurement bandwidth.

condition is not met, one must consider only the features of the measurement results and not their exact values.

4.3 Micro-Hall Devices

The ability to fabricate and manipulate the properties of magnetic structures at the mesoscale has not only led to the discovery of several new physical phenomena, but also to the development of a large pool of experimental techniques for studying such structures. In this regard, the micro-probe magnetometry based on the Hall effect is of particular interest. It offers high field sensitivity even when the dimensionality shrinks down to the nanoscale, has excellent operating ranges in terms of temperature and magnetic field, and it is non-invasive.

4.3.1 Technical Background

Recalling that in order to measure the conductivity σ of a semiconducting sample, when there are negligible gradients of carrier density, one has to take into account both the electrons concentrations n and mobility μ_e and the same quantities for holes, namely:

$$\sigma = |e|(n\mu_e + p\mu_h) = \frac{\vec{J}_{\mathrm{drift}}}{\vec{E}} \tag{4.2}$$

if—while a current is flowing along the $+x$ direction—a magnetic field is applied along the $+z$ direction, the drifting holes and electrons experience an average force in the same $(-y)$ direction, i.e.:

$$\vec{F}_e = +|e|\vec{E}\mu_e B_z \quad \text{for electrons} \tag{4.3}$$

$$\vec{F}_h = -|e|\vec{E}\mu_h B_z \quad \text{for holes} \tag{4.4}$$

Note that because the electron ($\vec{v_e} = \vec{E}\mu_e$) and hole velocities are in the opposite direction, the two forces are in the same direction. When these two forces are unequal, the surfaces along the $\pm y$ of the sample become charged and a potential difference arise between them. The appearence of this voltage is the Hall effect, and was discovered in 1879 by Edwin H. Hall.

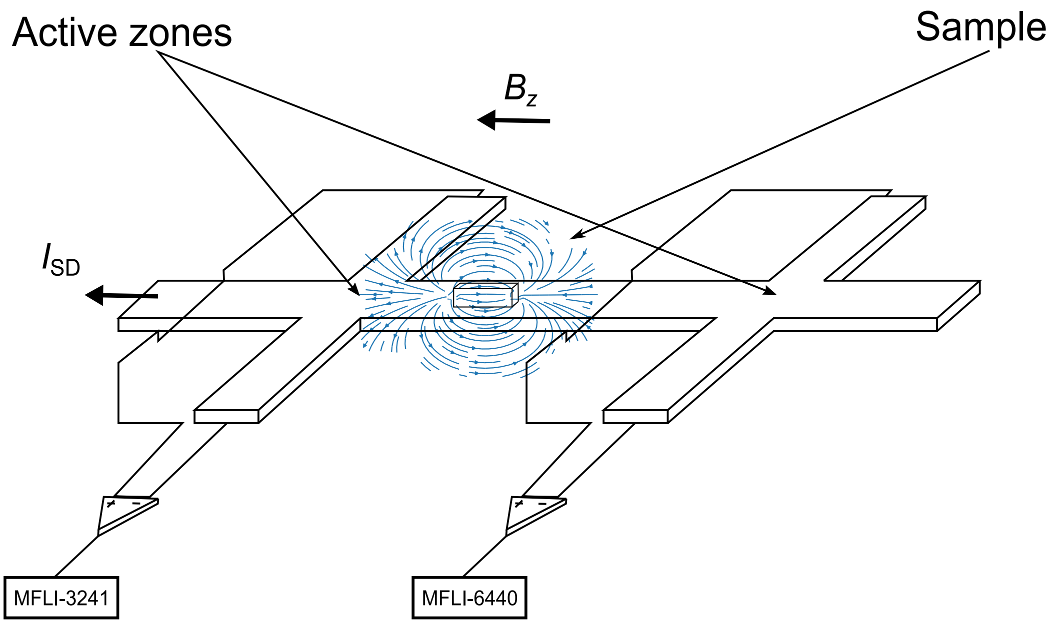

Figure 4.5: Cartoon of a micro-Hall sensor. Current flows along the central wires, while the sample sits on one of the two crosses. The sample is placed slightly offset compared to the crossposition as the magnetic field is less shielded by the gold surface. The signals are then measured in parallel by feeding first the voltage drops to two pre-amplifier and later to lock-in amplifiers.

The transverse electric field E_y builds up until the transverse force due to it becomes equal and opposite to the force due to the magnetic field. When this happens, the stationary condition $eE_y = ev_xB_z$ allows to define the Hall voltage as:

$$V_\mathrm{H} = wE_y = v_xB_z = \frac{j_xB_z}{ne} \tag{4.5}$$

where w in this case is the width of the semiconducting material, j_x is the component along x of current density and n is the charge carrier density. Considering that $j = I_\mathrm{SD}/wd$ where d is the specimen thickness, the expression may be rewritten as:

$$V_\mathrm{H} = R_\mathrm{H}I_\mathrm{SD}B_z \quad \text{with} \quad R_\mathrm{H} = \frac{E_y}{j_xB_z} = \frac{1}{ned} \tag{4.6}$$

The quantity R_H is known as Hall resistance and can be regarded by all means to the magnetic sensistivity of a particular probe. Clearly, the higher this values is the smaller field variation could be in principle detected. To be more specific, the minimum detectable magnetic field for a given area is the Hall voltage net of the electrical noise present when the circuit is closed. The noise is predominantly given by a Josephson and resistive (telegraph) noise.

Due to the pandemic of Covid-19 it has been impossible to find an adequate 2DEG substrate for this project. Nevertheless, I have tried to setup and detect the Hall signal from a Mn_12 microcrystals with a mesoscopic Hall made of Au. The Hall probe configuration with the simplest geometry is the Hall cross. When a magnetic object is placed close to the active area of the probe, the Hall signal will be proportional to the sum of the external field B and the magnetic stray field that enters the active area from the sample.

4.3.2 Fabrication and Experimental Setup

As previously observed in Sec 3.7, Cu powders and low-pass filters are used to filter DC lines extensively. Therefore, it is not possible to drive sufficient current through the wire spanning the two hall bars. In addition, when the current flowing through any load is low, the power of the Delft box module is significantly reduced†.

Devices used for magnetometry were fabricated by Dr Alex Gee (Fig. 4.8).

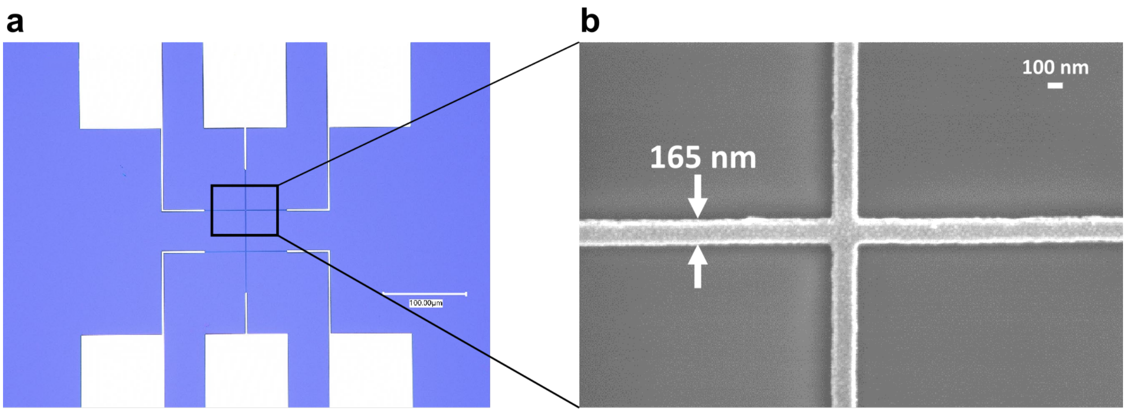

Figure 4.6: a, Optical microscope of a double-cross Hall bar fabricated with Au. The distance betwen the two crosses for this particular devices was 50 µm. **b,** Enlargment taken at the SEM of one of the two crosses, with the wire width displayed.

†If the impedance of the device under test (DUT) is low, the voltage drop across is also low since the current through the resistor is large. This causes the DAC's output voltage to drop dramatically.

4.3.3 Test Measurements

In order to prove the reliability of our apparatus and test its sensitivity, I have performed a few tests by measuring micro-crystals of Mn_{12}-acetate synthesised by Dr Dimitris Alexandropoulos. Mn_{12} is a molecular magnets whose magnetic response is well known in the literature.

The coercive field of Mn_{12} at cryogenic temperature is about 6 T, therefore I had to operate and temperatures higher for checking its magnetic response. In order to minimise chances the magnet would increase the temperature while was swept, I have slow sweep rates ranging from 0.83–1.67 mT s^{-1}.

We used SiO_2 as substrate where Hall devices with different spacing between the two crosses were used, ranging from 10 µm to 50 µm. The deposition and the correct orientation of the crystals reavealed to be not a negligible challenge and so we choose to use crystal approximately of 50 µm.

The transfer process was as followed. We first spread the Mn_{12} crystals on top of a filter paper in order to remove big chunck of crystals. In particular, molecular monemers present a needle-shape whereas dimers comes in a bulky-ball shape form. We then deposited a few of these crystals on top of a bare silicon substrate, where we also deposited a thin film of high-vacuum grease. We then lowered one of the probe in the probe station setup as described in Sec. 3.2 thanks to the micropositioner. Tungesten probes with a nominal diameter of 10 µm worked best for this purpose. We then gently deposited the grease within the two crosses paying attention specifically to not to damage the electrical connections.

The probe was then cleaned, and used to pick up a Mn_{12} crystal with the right lateral dimension. This was then gently lowered on top on the Hall bar device.

In order to maximise the dipolar field generated by the sample, the crystal should be perfectly oriented along the central gold pads. The difficulty of aligning the crystal in the desired direction is one of the weaknesses of our approach. Adjusting it by rotating the probe chuck is a viable alternative, but care has to be taken in not damaging the electrical connections. Another source of problem is that often the crystal break upon touching the substrate. Although monemer of Mn_{12} should be relatively robust against the forces at play in this scenario, breaks might result as due to grains of crystals attached to each other.

4.3.4 Current Measurement Test of the Hall Effect

The first test I did was to make sure that the Hall effect was present at varying measurement bias. From Eq. 4.6 the Hall voltage V_H is proportional to the current and the magnetic field and inversly proportional to the carrier density n of the conductor. As shown in Fig. 4.7, I have used only one lockin amplifier since only we were only intersted in the voltage built up across just one cross. The signal was then fed into a lock-in, and the digitise signal was then read directly in a standard desktop via an usb cable.

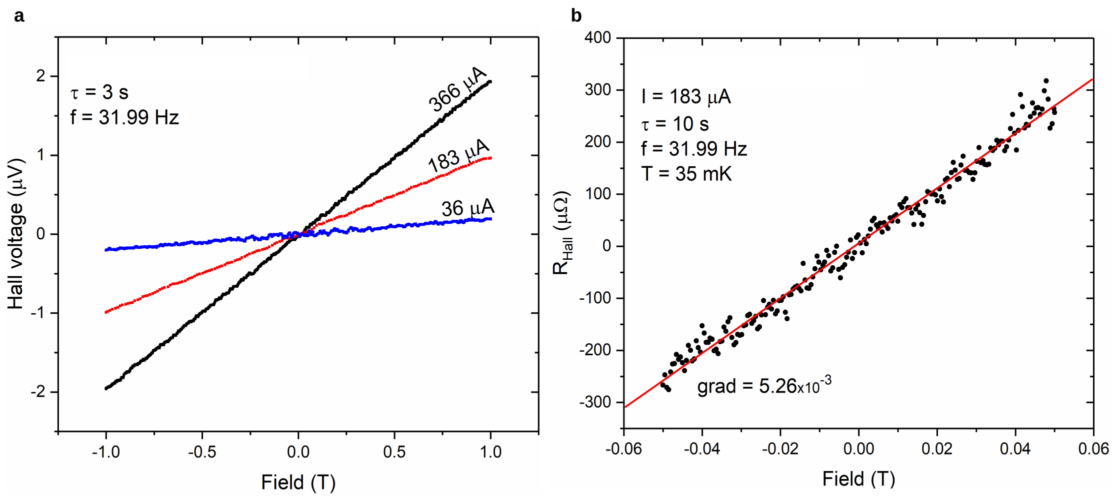

Figure 4.7: a, Hall voltage measured across one cross when the current driven is increased. The signal is proportional to the current as expected. The parameters f and τ are the lock-in operation frequency and integration time, respectively. The frequency in particular was optimised by taking a frequency scan the line response and set it to the point of minimum noise. **b,** The Hall coefficient for this particular device was calcualted to be $R_H \sim 1 \times 10^{-6}\,\Omega G^{-1}$ which is expected for an Au probe of $d \sim 1\,\mu m$

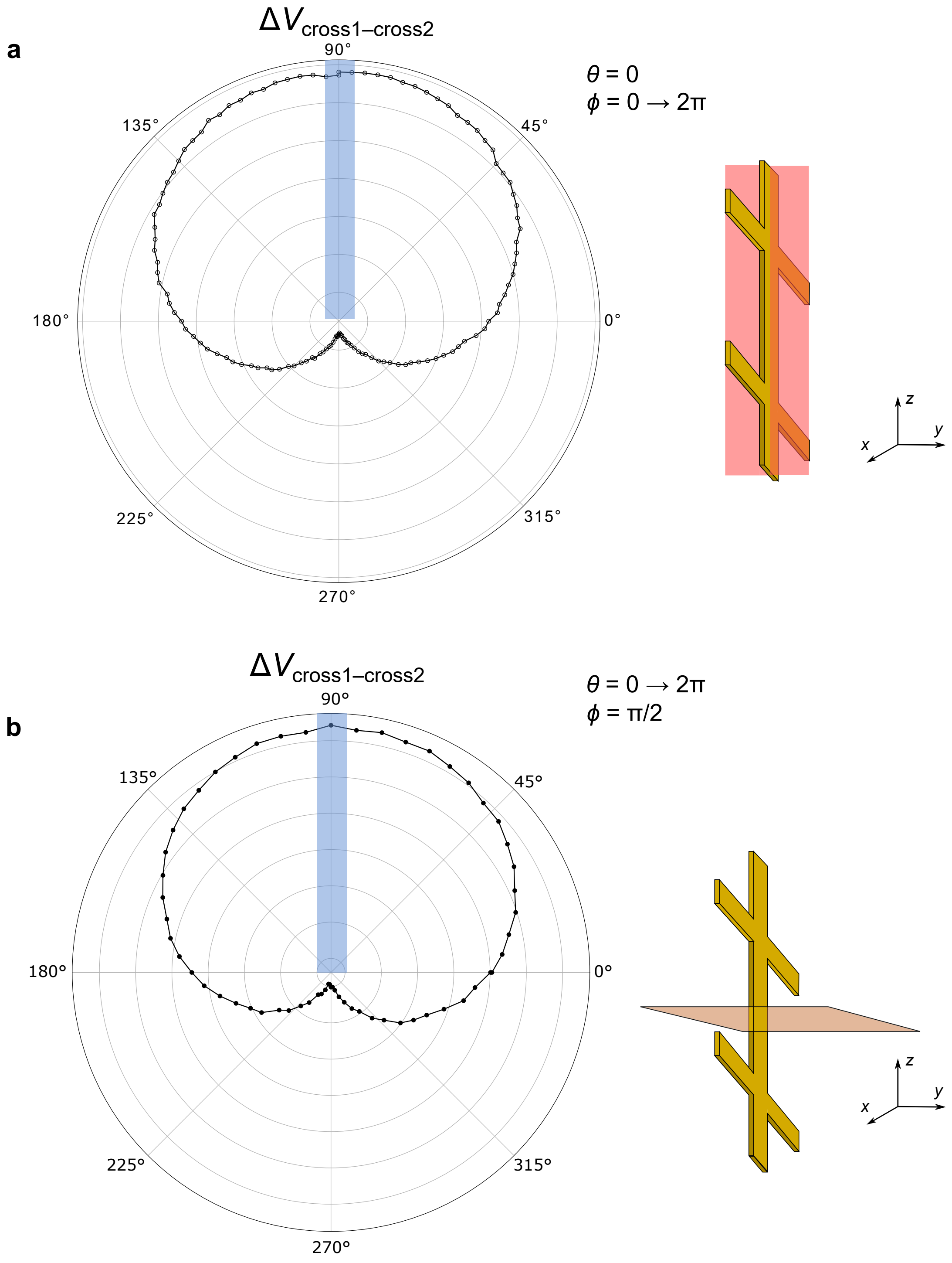

Figure 4.8: a, The difference between the voltages build up across the two crosses was recoded thorugh angle-resolved experiments. The direction of maximum signal is highlighted by the blue rectangle and along the y axis, as expected from the Hall effect. On the right, a schematic along which normal-plane the magnet was rotated along. θ and ϕ are the sperhical parameters to input in the routine I wrote. **b,** polar plot of the response for a x-oriented normal-plane. Once again, the region of maximum signal is found when the magnetic field component is perpendicular to the surface of the two crosses.

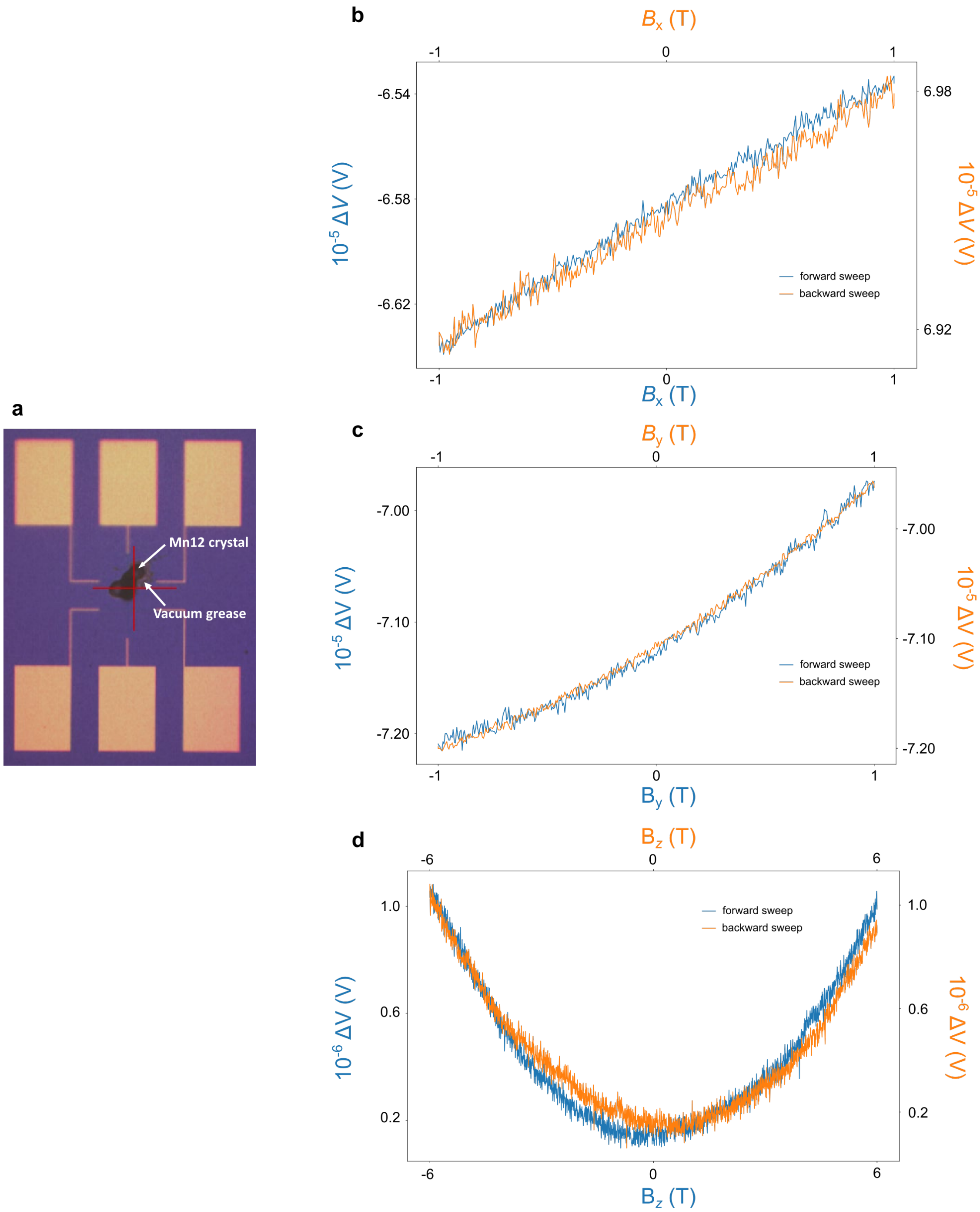

Figure 4.9: Cartoon of a micro-Hall sensor. Current flows along the central wires, while the sample sits on one of the two crosses. The sample is placed slightly offset compared to the crossposition as the magnetic field is less shielded by the gold surface. The signals are then measured in parallel by feeding first the voltage drops to two pre-amplifier and later to lock-in amplifiers.

4.3.5 Anisotropy Test

4.4 Graphene Hall Bars

The low sensistivity achieved with the Hall devices detailed in the preceeding chapter depends, as we have concluded, by the high charge carrier concentration n experiencing the Hall effect induced by the induction field placed along the y axis. Therefore, the Hall effect induced by the dipolar field of the sample when the induction field is oriented along z is too weak, resulting in a too small voltage drop between the two crosses even using AC techiniques.

A workaround to this situation would be to use a material where the carrier concentration is minimum, such that the $1/n$ term in Eq. 4.6 is large enough to make the Hall coefficient measurable. Such materials are, for example, 2DEGs where concentrations $n \simeq 3 \times 10^{-11}\,\mathrm{cm}^{-2}$ and mobilities $\mu \simeq 1 \times 10^{6}\,\mathrm{cm\,V^{-2}\,s^{-1}}$ offer an ideal platform to detect the Hall effect produced by the dipolar field of a magnet placed on top of the device. Unfortunately, the recent semiconductor crisis and the scarce availability of such substrates have hampered the realisation of such device in house.

Another suitable material for Hall probe is graphene. Thanks to its outstanding mobilities values and the possibility to tune the carrier density by gating the Fermi level, several architecture probes ranging from the epitaxially-grown graphene to fully encapsulated devices in hBN. More challening has been proved to realised an adequate device by adopting CVD-grown graphene, as it is notoriously difficult to achieve good Hall coefficient since it depends, in first approximation, linearly with the eleastic mean free path $R_\mathrm{H} \simeq \ell_\mathrm{el}$.

4.4.1 Fabrication

Graphene Hall bar (GHB) devices were fabricated by lithography means, reusing a typical two-terminal transistor architecture on 300 nm SiO_2 with n-doped Si as backgate. The fabrication procedure begins with the removal of PMMA in acetone overnight, followed by a rinse in IPA and ebam exposure of negative resist ARN7500